

LEVEL-VI

SINGLE ANSWER QUESTIONS

1. The equations of the lines through $(-1, -1)$ and making angle 45° with the line $x + y = 0$ are given by

A) $x^2 - xy + x - y = 0$ B) $xy - y^2 + x - y = 0$
 C) $xy + x + y = 0$ D) $xy + x + y + 1 = 0$

2. The ends of the base of an isosceles triangle are at $(2, 0)$ and $(0, 1)$ and the equation of one side is $x = 2$ then the orthocentre of the triangle is

A) $\left(\frac{3}{2}, \frac{3}{2}\right)$ B) $\left(\frac{5}{4}, 1\right)$
 C) $\left(\frac{3}{4}, 1\right)$ D) $\left(\frac{4}{3}, \frac{7}{12}\right)$

3. If d_1, d_2, d_3 be perpendiculars from the points $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$ respectively on the line

$$x \cos \alpha + y \sin \alpha + \frac{\sin^2 \alpha}{\cos \alpha} = 0 \text{ with } x_1, x_2, x_3$$

are in G.P. and y_1, y_2, y_3 are in A.P. and A, C lie on the curve $y^2 = 4x$ then d_1, d_2, d_3 are in

A) A.P. B) G.P. C) H.P. D) A.G.P

4. The equation of the pair of the lines passing through the origin and having slope $m \in I$ for which equation $(x - 3)(x + m) + 1 = 0$ has integral roots is

A) $y^2 - 6xy - 5x^2 = 0$
 B) $y^2 - 6xy + 5x^2 = 0$
 C) $y^2 + 6xy - 5x^2 = 0$
 D) $y^2 + 6xy + 5x^2 = 0$

5. The line $x + y = 1$ meets the lines represented by the equation $y^3 - 6xy^2 + 11x^2y - 6x^3 = 0$ at the points P, Q, R. If O is the origin, then $(OP)^2 + (OQ)^2 + (OR)^2$ is equal to

A) $\frac{87}{72}$ B) $\frac{121}{72}$ C) $\frac{211}{72}$ D) $\frac{217}{72}$

6. Let $f(x + y) = f(x) \cdot f(y) \quad \forall x, y \in \mathbb{R}$, $f(1) = 2$. Area enclosed by $3|x| + 2|y| \leq 8$ is $\frac{f(k)}{m}$. The point P $(2, 6)$ is translated parallel to $y = mx$ in the first quadrant through a 'k' unit distance. The coordinates of the new position of P is/are

A) $\left(2 \pm \frac{2}{\sqrt{10}}, 6 \pm \frac{1}{\sqrt{10}}\right)$
 B) $\left(2 \pm \frac{2}{\sqrt{10}}, 6 \pm \frac{18}{\sqrt{10}}\right)$
 C) $\left(2 \pm \frac{6}{\sqrt{10}}, 6 \pm \frac{18}{\sqrt{10}}\right)$
 D) $\left(2 \pm \frac{6}{\sqrt{10}}, 6 + \frac{18}{\sqrt{10}}\right)$

7. If $P = \left(\frac{1}{x_p}, p\right); Q = \left(\frac{1}{x_q}, q\right); R = \left(\frac{1}{x_r}, r\right)$

where $x_k \neq 0$, denotes the k^{th} terms of H.P. for $k \in N$, then

A) area of

$$\Delta PQR = \left(\frac{p^2 q^2 r^2}{2} \sqrt{(p-q)^2 + (q-r)^2 + (r-p)^2} \right)$$

B) ΔPQR is a right angled triangle.

C) The points P, Q, R are collinear

D) None of these.

8. If the lines $x + y + 1 = 0; 4x + 3y + 4 = 0$ and $x + \alpha y + \beta = 0$, where $\alpha^2 + \beta^2 = 2$, are concurrent then

A) $\alpha = 1, \beta = -1$ B) $\alpha = 1, \beta = \pm 1$
 C) $\alpha = -1, \beta = \pm 1$ D) $\alpha = -1, \beta = 1$

9. Let $0 < \theta < \frac{\pi}{2}$ be a fixed angle. If

$P(\cos \theta, \sin \theta)$ and $Q(\cos(\alpha - \theta), \sin(\alpha - \theta))$

then Q is obtained from P by [IIT 2002]

A) clockwise rotation around origin through an angle α

- B) anticlockwise rotation around origin through an angle α
 C) reflection in the line through origin with slope $\tan \alpha$
 D) reflection in the line through origin with slope $\tan \frac{\alpha}{2}$
10. The locus of the orthocenter of the triangle formed by the lines
 $(1+p)x - py + p(1+p) = 0$
 $(1+q)x - qy + q(1+q) = 0$
 and $y = 0$, where $p \neq q$, is [IIT 2009]
 A) a hyperbola B) a parabola
 C) an ellipse D) a straight line
11. Vertex A of $\triangle ABC$ moves in such way that $\tan B + \tan C = a$ constant, when BC is fixed then locus of orthocentre of $\triangle ABC$ is a
 A) Straight line B) Parabola
 C) Ellipse D) Circle
12. Two vertices of a Triangle are (1,3) and (4,7). the orthocentre lies on the line $x+y=3$. the locus of third vertex is
 A) $x^2 - 2xy + 2y^2 - 3x - 4y + 36 = 0$
 B) $2x^2 - 4xy + 3y^2 - 4x - y + 42 = 0$
 C) $3x^2 + xy - 4y^2 - 2x - 24y - 40 = 0$
 D) $x^2 - 4xy + 3y^2 - 2x - y - 40 = 0$
13. If the line $ax+by=1$ passes through point of intersection of $y = x \tan \alpha + p \sec \alpha$;
 $y \sin(30^\circ - \alpha) - x \cos(30^\circ - \alpha) = p$ and is inclined at 30° with $y = \tan \alpha$, then
 $a^2 + b^2 =$
 A) $\frac{1}{p^2}$ B) $\frac{2}{p^2}$ C) $\frac{3}{2p^2}$ D) $\frac{3}{4p^2}$
14. Let $A(6,7), B(2,3), C(-2,1)$ vertices of a Triangle. The point P in the interior of $\triangle ABC$ such that $\triangle PBC$ is an equilateral triangle is
 A) $(-\sqrt{3}, 2+2\sqrt{3})$ B) $(-\sqrt{3}, 2-2\sqrt{3})$
 C) $(\sqrt{3}, 2-2\sqrt{3})$ D) $(\sqrt{3}, 2+2\sqrt{3})$
15. In an Equilateral triangle ex-centre opposite vertex A is (2, -4) and equation of side BC is $x+y-2=0$. If coordinates of vertex A is (α, β) then $2\alpha - \beta$
 A) 12 B) 6 C) -6 D) 4
16. A variable line whose slope is -2 cuts x and y axes respectively at points A and C. A Rhombus ABCD is completed. such that the vertex B lies on the line $y = x$, then the locus of the vertex D is
 A) $x+y+2=0$ B) $x+y+3=0$
 C) $x+y=0$ D) $x+y-1=0$
17. The length of line segment joining the feet of the perpendiculars drawn from the point (3,4) on the pair of lines $x^2 - 5xy + 6y^2 = 0$ is
 A) $\sqrt{2}$ B) 2 C) $\frac{1}{\sqrt{2}}$ D) $2\sqrt{2}$
18. The number of possible straight lines, passing through (2,3) and forming a triangle with coordinate axes, whose area is 12 sq. units, is :
 A) 1 B) 2 C) 3 D) 4
19. Given A (0, 0) and B (x, y) with $x \in (0, 1)$ and $y > 0$. Let the slope of the line AB equals m_1 . Point C lies on the line $x=1$ such that the slope of BC equals m_2 where $0 < m_2 < m_1$. If the area of the triangle ABC can be expressed as $(m_1 - m_2)f(x)$, then the largest possible value of $f(x)$ is :
 A) 1 B) $1/2$ C) $1/4$ D) $1/8$

MULTI ANSWER QUESTIONS

20. If one diagonal of a square is the portion of line $\frac{x}{a} + \frac{y}{b} = 1$ intercepted by the axes, then the extremities of the other diagonal of the square are
 A) $\left(\frac{a+b}{2}, \frac{a+b}{2}\right)$ B) $\left(\frac{a-b}{2}, \frac{a+b}{2}\right)$
 C) $\left(\frac{a-b}{2}, \frac{b-a}{2}\right)$ D) $\left(\frac{a+b}{2}, \frac{b-a}{2}\right)$
21. If one vertex of an equilateral triangle of side 'a' lies at origin and other lies on the

STRAIGHT LINES

line $x - \sqrt{3}y = 0$, then the coordinates of third vertex are

- A) $(0, a)$ B) $\left(\frac{\sqrt{3}}{2}a, \frac{-a}{2}\right)$
 C) $(0, -a)$ D) $\left(\frac{-\sqrt{3}}{2}a, \frac{a}{2}\right)$

22. Let B $(1, -3)$ and D $(0, 4)$ represents two vertices of a rhombus ABCD in xy plane, then coordinates of vertex A if $\angle BAD = 60^\circ$ can be equal to:

A) $\left(\frac{1-7\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}\right)$ B) $\left(\frac{1+7\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$

C) $\left(\frac{-1+7\sqrt{3}}{2}, \frac{-1+\sqrt{3}}{2}\right)$ D) $\left(\frac{-1-7\sqrt{3}}{2}, \frac{-1-\sqrt{3}}{2}\right)$

23. If $6a^2 + 12b^2 + 2c^2 + 17ab - 10bc - 7ac = 0$ then all the lines represented by $ax + by + c = 0$ are concurrent at the point

A) $(-2, -3)$ B) $(2, 3)$

C) $\left(-\frac{3}{2}, -2\right)$ D) $\left(-\frac{3}{2}, 2\right)$

24. The point A $(0, 0)$, B $(\cos \alpha, \sin \alpha)$ and C $(\cos \beta, \sin \beta)$ are the vertices of a right angled triangle if;

A) $\sin\left(\frac{\alpha-\beta}{2}\right) = \frac{1}{\sqrt{2}}$ B) $\cos\left(\frac{\alpha-\beta}{2}\right) = -\frac{1}{\sqrt{2}}$

C) $\cos\left(\frac{\alpha-\beta}{2}\right) = \frac{1}{\sqrt{2}}$ D) $\sin\left(\frac{\alpha-\beta}{2}\right) = -\frac{1}{\sqrt{2}}$

25. Let A (x_1, y_1) , B (x_2, y_2) , C (x_3, y_3) be the vertices of the triangle ABC such that $x_3 = lx_1 + mx_2$, $y_3 = ly_1 + my_2$ where

$l, m < 0$ then, origin

- A) Lies out side triangle ABC

B) And C lies on the same side of AB

C) May lie inside triangle ABC

D) None of these

26. Two sides of a rhombus OABC

(lying entirely in Q_1 or Q_3) of area equal to

2 square units are $y = \frac{x}{\sqrt{3}}$, $y = \sqrt{3}x$.

The possible coordinates of B is / are ('O' is origin)

A) $(1 + \sqrt{3}, 1 + \sqrt{3})$ B) $(-1 - \sqrt{3}, -1 - \sqrt{3})$

C) $(\sqrt{3} - 1, \sqrt{3} - 1)$ D) $(1 - \sqrt{3}, 1 - \sqrt{3})$

COMPREHENSION QUESTIONS

PASSAGE : I

A $(1, 3)$ and C $\left(-\frac{2}{5}, -\frac{2}{5}\right)$ are the vertices of a triangle ABC and the equation of the angle bisector of $\angle ABC$ is $x + y = 2$

27. Equation of BC is

A) $7x + 3y + 4 = 0$ B) $3x + 7y + 4 = 0$

C) $13x + 7y + 8 = 0$ D) $x + 9y + 4 = 0$

28. Coordinates of vertex B

A) $\left(\frac{3}{10}, \frac{17}{10}\right)$ B) $\left(\frac{17}{10}, \frac{3}{10}\right)$

C) $\left(-\frac{5}{2}, \frac{9}{2}\right)$ D) $(1, 1)$

29. Equation of side AB is

A) $13x - 7y + 8 = 0$ B) $13x + 7y - 34 = 0$

C) $3x + 7y - 24 = 0$ D) $3x + 7y + 24 = 0$

PASSAGE : II

The base of an isosceles triangle is equal to 4 units, the base angle is equal to 45° .

A straight line cuts the extension of the base at a point M at the angle θ and bisects the lateral side of the triangle which is nearest to M.

30. The area of quadrilateral which the straight line cuts off from the given triangle is

$$\text{A) } \frac{3 + \tan \theta}{1 + \tan \theta} \quad \text{B) } \frac{3 + 2 \tan \theta}{1 + \tan \theta}$$

$$\text{C) } \frac{3 + \tan \theta}{1 - \tan \theta} \quad \text{D) } \frac{3 + 5 \tan \theta}{1 + \tan \theta}$$

31. The possible range of values in which area of quadrilateral which straight line cuts off from the given triangle lie in

$$\text{A) } \left(\frac{5}{2}, \frac{7}{2}\right) \quad \text{B) } (4, 5) \quad \text{C) } \left(4, \frac{9}{2}\right) \quad \text{D) } (3, 4)$$

32. The length of portion of straight line inside the triangle may lie in the range

$$\text{A) } (2, 4) \quad \text{B) } \left(\frac{3}{2}, \sqrt{3}\right) \quad \text{C) } (\sqrt{2}, 2) \quad \text{D) } (\sqrt{2}, \sqrt{3})$$

PASSAGE : III

Given point A (6, 30) and point B (24, 6) equation of line AB is $4x + 3y = 114$. Point P (0, λ) is a point on y - axis such that $0 < \lambda < 38$ and point Q(0, k) is a point on y-axis such that $k > 38$.

33. For all positions of point P, angle APB is maximum when point P is

$$\text{A) } (0, 12) \quad \text{B) } (0, 15) \quad \text{C) } (0, 18) \quad \text{D) } (0, 21)$$

34. The maximum value of angle APB is

$$\text{A) } \frac{\pi}{3} \quad \text{B) } \frac{\pi}{2} \quad \text{C) } \frac{2\pi}{3} \quad \text{D) } \frac{3\pi}{4}$$

35. For all position of point Q, angle AQB is maximum when point Q is:

$$\text{a) } (0, 54) \quad \text{b) } (0, 58) \quad \text{c) } (0, 60) \quad \text{d) } (0, 1)$$

PASSAGE : IV

Consider a variable line 'L' which passes through the point of intersection P of the lines $3x + 4y - 12 = 0$ and $x + 2y - 5 = 0$ meeting the coordinate axes at point A and B.

36. Locus of the middle point of the segment AB has the equation

$$\text{A) } 3x + 4y = 4xy \quad \text{B) } 3x + 4y = 3xy$$

$$\text{C) } 4x + 3y = 4xy \quad \text{D) } 4x + 3y = 3xy$$

37. Locus of the feet of the perpendicular from the origin on the variable line L has the equation

$$\text{A) } 2(x^2 + y^2) - 3x - 4y = 0$$

$$\text{B) } 2(x^2 + y^2) - 4x - 3y = 0$$

$$\text{C) } x^2 + y^2 - 3x - y = 0$$

$$\text{D) } x^2 + y^2 - x - 2y = 0$$

38. Locus of the centroid of the variable triangle OAB has the equation (where O is origin)

$$\text{a) } 3x + 4y + 6xy = 0 \quad \text{b) } 4x + 3y - 6xy = 0$$

$$\text{c) } 3x + 4y - 6xy = 0 \quad \text{d) } 4x + 3y + 6xy = 0$$

PASSAGE : V

Let a and b be the lengths of the legs of a right triangle with following properties

- (i) All 3 sides of the triangle are integers
(ii) the perimeter of the triangle is numerically equal to area of the triangle, it is given that $a < b$

39. The number of ordered pairs (a, b) will be

$$\text{a) } 1 \quad \text{b) } 2 \quad \text{c) } 3 \quad \text{d) } 4$$

40. Maximum possible perimeter of the triangle is

$$\text{a) } 27 \quad \text{b) } 28 \quad \text{c) } 29 \quad \text{d) } 30$$

41. Minimum possible area of the triangle is

$$\text{a) } 24 \quad \text{b) } 25 \quad \text{c) } 26 \quad \text{d) } 27$$

PASSAGE : VI

Consider three non collinear points A(9, 3), B(7, -1) and C(1, -1). Let P(a, b) be the centre and R is the radius of circle 'S' passing through points A, B, C. Also $H(\bar{x}, \bar{y})$ are the coordinates of the orthocentre of triangle ABC whose area be denoted by Δ .

42. If D, E, F are the middle points of BC, CA and AB respectively then the area of the triangle DEF is

$$\text{A) } 12 \quad \text{B) } 6 \quad \text{C) } 4 \quad \text{D) } 3$$

43. The value of $a + b + R$ is

$$\text{a) } 3 \quad \text{b) } 12 \quad \text{c) } 13 \quad \text{d) } \text{None}$$

44. The ordered pair $(\bar{x}, \bar{y})$ is

$$\text{a) } (9, 6) \quad \text{b) } (-9, 6) \quad \text{c) } (9, -5) \quad \text{d) } (9, 5)$$

PASSAGE : VII

An equilateral triangle ABC has its centroid at the origin and one side 'AC' is $x + y = 1$

45. If $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$ be the vertices of the equilateral triangle. Then

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$$x_1x_2 + x_2x_3 + x_3x_1 =$$

- A) -1 B) -3/2
C) 1 D) none of these

46. A point P is taken on the minor arc AC of the circumcircle of triangle ABC. Then maximum value of (PA+PB+PC) is equal to

- A) $4\sqrt{2}$ B) $\sqrt{2}$ C) $3\sqrt{2}$ D) $9/\sqrt{2}$

47. Another point 'P' is taken in the plane of ABCP. (P being PA+PB+PC is maximum).

If $P^1A + P^1B + P^1C + P^1P = \sqrt{6} + \sqrt{8}$, then locus of P^1 is

- A) square B) circle
C) a point D) none of these

PASSAGE : VIII

Let the lines represented by the equation $x^2y^2 - x^2 - y^2 + 1 = 0$ form a square ABCD. A point 'P' is taken on the same plane of square such that atleast two sides of all the triangles PAB, PBC, PCD and PDA are equal.

48. The number of possible position of the point 'P'

- A) 1 B) 5 C) 17 D) 9

49. For all the possible position of point 'P'. The number of the given triangles have all the three sides equal

- A) 0 B) 4 C) 8 D) 5

50. One of the possible position of the 'P' such that atleast one of the given triangle is equilateral is given by

- A) (0, 0) B) $(\sqrt{3}+1, 0)$
C) $(0, \sqrt{3}-2)$ D) $(\sqrt{3}+1, \sqrt{3}-1)$

PASSAGE : IX

Consider $\triangle ABC$ with incentre $I(1, 0)$.

Equations of the straight lines AI, BI, CI are $x=1, y+1=x$ and $x+3y=1$ respectively and

$$\cot \frac{A}{2} = 2$$

51. Equation to the locus of centroid of $\triangle ABC$ is

- A) $x=8y+1$ B) $x=2y+1$
C) $2x-3y=1$ D) $3x=5y$

52. Slope of side BC is

- A) $\frac{1}{2}$ B) $-\frac{1}{3}$ C) $\frac{1}{4}$ D) $\frac{2}{3}$

53. If point A lies above the x-axis and area of

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$\triangle ABC$ is 30 sq. units, then the in radius of $\triangle ABC$ is

- A) $\sqrt{3}$ B) $\sqrt{5}$ C) 2 D) $3\sqrt{2}$

MATRIX MATCHING QUESTIONS

54. Match the following
Column I

A) Lines $x-2y-6=0, 3x+y-4=0$ and $\lambda x+4y+\lambda^2=0$ are concurrent. The value of λ is

B) The points $(\lambda+1, 1), (2\lambda+1, 3)$ and $(2\lambda+2, 2\lambda)$ are collinear, then the value of λ is

C) If line $x+y-1-\lambda=0$, passing through the intersections of $x-y+1=0$ and $3x+y-5=0$ is perpendicular to one of them then the value of λ is

D) If the line $y-x-1+\lambda=0$ is equally inclined to axes and equidistant from the points (1, -2) and (3, 4) then λ is

Column II

p) 2

q) -4

r) $-\frac{1}{2}$

s) 1

55. Reflection of the line $x+y+1=0$ in the line.

Column I

A) $2x+y+1=0$

B) $x-2y+1=0$

C) $x+2y-1=0$

D) $2x+y-1=0$

Column II

p) $x+7y-11=0$

q) $7x+y+1=0$

r) $7x+y-11=0$

s) $7x+y+7=0$

56. Vertex A of the $\triangle ABC$ is at origin. The equation of medians through B and C are $15x-4y-240=0$ and $15x-52y+240=0$ respectively.

Column I

a) The coordinates of

incentre of $\triangle ABC$ are

b) The coordinates of

centroid of $\triangle ABC$ are

c) The coordinates of

excenter opposite to vertex r) (12, 21)

Column II

p) $\left(\frac{56}{3}, 10\right)$

q) (21, 12)

C of $\triangle ABC$ are

d) The coordinates of orthocentre of $\triangle ABC$ are

s) $(-4, 7)$

t) $(0, 63)$

57. Consider the line given by

$$L_1 = x + 3y - 5 = 0, L_2 = 3x - ky - 1 = 0,$$

match the statements / expressions in Column-I with the statements / expressions in Column-II. [IIT-JEE 2008]

a) L_1, L_2, L_3 are concurrent, if p) $k = -9$

b) One L_1, L_2, L_3 is parallel to at least one of the other two, q) $k = -6/5$ if

c) L_1, L_2, L_3 form a triangle, if r) $k = 5/6$

D) L_1, L_2, L_3 do not form a triangle, if s) $k = 5$

58. In triangle ABC the equation of side BC is $2x - y = 3$ and its circumcentre and orthocentre are at $(2, 4)$ and $(1, 2)$ respectively then match the items of column I with items of column II

Column I

Column II

(a) $\tan B \tan C$ (p) $\frac{4\sqrt{13}}{\sqrt{5}}$

(b) The length of BC (q) $\frac{18\sqrt{13}}{5}$

(c) $\sin A$ (r) 3

(d) Area of the triangle is (s) $\frac{2\sqrt{13}}{\sqrt{61}}$

INTEGER QUESTIONS

59. If the slope of one of the line given by $36x^2 + 2hxy + 72y^2 = 0$ is four times the other, value of $h^2 - 4045$ is

60. A line L through A $(2, 5)$ intersects the lines $2x^2 - 5xy + 2y^2 = 0$ at P and Q such that for some point R on L; AP, AR, AQ are in H.P. If the locus of R is $\lambda x - 10y = 0$ then $\frac{\lambda + 1}{6} =$

61. The sum of algebraic distances from the points (n, n^2) where $n = 1, 2, 3, \dots, 2013$, to a straight line is zero and if such a line always passes through the point (α, β) and

$$\frac{1}{\alpha} \left(\beta - \frac{1007}{3} \right) = abcd \text{ then}$$

$b + d =$ (abcd is four digit number)

62. The straight line $L \equiv x + y + 1 = 0$ and $L_1 \equiv x + 2y + 3 = 0$ are intersecting, m is the slope of the straight line L_2 such that L is the bisector of the angle between L_1 and L_2 . The value of m^2 is.

63. If the straight lines joining the origin to the point of intersection of the straight line $69x + 25y = 3450$ and the circle $(x - 25)^2 + (y - 69)^2 = c^2$ are at right angle then $c^2 - 5380$ is

64. If $ax^2 - 2hxy - ay^2 = 0$ represents a pair of straight lines forming with $2x + 3y = 6$ an isosceles right angled triangle at the origin then $|-a + h|$ is.

65. Two sides of a rhombus lying in the first quadrant are given by $3x - 4y = 0$ and $12x - 5y = 0$. The length of the longer diagonal is 12. If the equations of other two

sides are $3x - 4y + \frac{180}{\sqrt{a}} = 0$ and

$$12x - 5y = \frac{468}{\sqrt{a}} \text{ then } \frac{a - 40}{10} =$$

66. Let the sides of triangle ABC are all integers with A as origin. If $(2, -1)$ and $(3, 6)$ are points on the line AB and AC respectively (lines AB and AC may be extended to contain these points) and lengths of any two sides are primes that differ by 50. If a is least possible length of the third side and S is least possible perimeter of the triangle and $a \leq S \leq 1930$ then N = ...

67. Consider two lines $L_1 = x - y = 0$ and $L_2 = x + y = 0$ and a moving point P (

x, y). Let $d(P, L_i), i=1, 2$ represents the perpendicular distance of the point P from L_i . If point P moves in certain region R in

such a way that $\sum_{i=1}^2 d(P, L_i) \in [2, 4]$, Let

the area of region R is A, then find $\frac{A}{4}$.

68. The Coordinates of the feet of the perpendicular from the vertices of a triangle on the opposite sides are D(20, 25), E(8, 16), F(8, 9). The number of such triangles are
69. Number of Equilateral triangles that can be formed with points lying on the curve $8x^3 + y^3 + 6xy = 1$

KEY

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SINGLE ANSWER QUESTIONS

- 1) D 2) B 3) B 4) D
5) B 6) C 7) C 8) D
9) D 10) D 11) A 12) C
13) D 14) A 15) A 16) C
17) C 18) C 19) D

MULTIPLE ANSWER QUESTIONS

- 20) A, C 21) A, B, C, D 22) A, B
23) A, C 24) A, B, C, D 25) B, C
26) A, B

COMPREHENSION QUESTIONS

- 27) A 28) C 29) C 30) D
31) D 32) C 33) C 34) B
35) B 36) A 37) B 38) C
39) B 40) D 41) A 42) D
43) B 44) C 45) B 46) A
47) C 48) D 49) C 50) B
51) B 52) A 53) B

MATRIX MATCH QUESTIONS

- 54) A-p, q; B-p, r; C-p; D-p
55) A-q; B-s; C-p; D-r
56) A-q; B-p; C-s; D-t
57) A-s; B-p, q; C-r; D-p, q, s
58) A-r, B-p, C-s, D-q

INTEGER QUESTIONS

- 59) 5 60) 3 61) 5 62) 4
63) 6 64) 7 65) 9 66) 4
67) 6 68) 4 69) 1

HINTS
LEVEL - VI

SINGLE ANSWER QUESTIONS

1. The lines will be parallel to coordinate axes given by $(x+1)(y+1) = 0$

$$\Rightarrow xy + x + y + 1 = 0$$

2. Let the orthocentre be $(\lambda, 1)$ then

$$\frac{\lambda - \frac{1}{2}}{\lambda - 1} = 2 \Rightarrow \lambda = \frac{5}{4}$$

$$\therefore \text{Orthocentre} = \left(\frac{5}{4}, 1\right)$$

- 3.

$$d_1 = \left| x_2 \cos \alpha + y_1 \sin \alpha + \frac{\sin^2 \alpha}{\cos \alpha} \right|$$

$$d_2 = \left| x_2 \cos \alpha + y_2 \sin \alpha + \frac{\sin^2 \alpha}{\cos \alpha} \right|,$$

$$d_3 = \left| x_3 \cos \alpha + y_3 \sin \alpha + \frac{\sin^2 \alpha}{\cos \alpha} \right|$$

$$x_2^2 = x_1 x_3 \text{ and } 2y_2 = y_1 + y_3$$

$$\text{Clearly, } d_2^2 = d_1 d_3$$

4. $(x-3)(x-m)+1=0$

$$\Rightarrow x^2 + (m-3)x + (1-3m) = 0$$

$$\therefore x = \frac{-(m-3) \pm \sqrt{(m-3)^2 - 4(1-3m)}}{2}$$

$$= \frac{(m-3) \pm \sqrt{m^2 + 6m + 5}}{2}$$

$$x \text{ is integer} \Rightarrow m^2 + 6m + 5 = 0$$

$$\Rightarrow m = -1, -5$$

Required pair of equation is

$$(y+x)(y+5x) = 0$$

$$\text{i.e., } y^2 + 6xy + 5x^2 = 0$$

5. $y^3 - 6xy^2 + 11x^2y - 6x^3 = 0$
 By inspection $y = x$ is a solution
 other equations are $y = 2x, y = 3x$
 Given equation $x + y = 1$ (1)
 P = point of intersection of (1) and $y - x = 0$

$$= \left(\frac{1}{2}, \frac{1}{2}\right)$$

Q = point of intersection (1) & $y - 2x = 0$

$$= \left(\frac{1}{3}, \frac{2}{3}\right)$$

R = point of intersection of (1) & $y - 3x = 0$

$$= \left(\frac{1}{4}, \frac{3}{4}\right)$$

$$\therefore (OP)^2 + (OQ)^2 + (OR)^2$$

$$= \frac{1}{4} + \frac{1}{4} + \frac{1}{9} + \frac{4}{9} + \frac{1}{16} + \frac{9}{16}$$

$$= \frac{121}{72}$$

6. $f(x) = 2^x$, Area of region $= 4 \left(\frac{64}{2(3)(2)} \right) = \frac{64}{3}$

Here $K = 6, m = 3$

Slope of line $y = mx$ is $\tan \theta = 3$

$$\cos \theta = \frac{1}{\sqrt{10}}, \sin \theta = \frac{3}{\sqrt{10}}$$

$$\text{Required points} = \left(2 \pm \frac{6}{\sqrt{10}}, 6 \pm \frac{18}{\sqrt{10}} \right)$$

$$7. \Rightarrow \begin{vmatrix} a + (p-1)d & p & 1 \\ a + (q-1)d & q & 1 \\ a + (r-1)d & r & 1 \end{vmatrix} = 0$$

$$8. \begin{vmatrix} 1 & 1 & 1 \\ 4 & 3 & 4 \\ 1 & \alpha & \beta \end{vmatrix} = 0 \Rightarrow \beta = 1$$

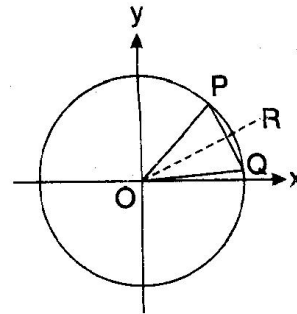
$$\alpha^2 + \beta^2 = 2 \Rightarrow \alpha = -1$$

9. The points
 $P(\cos \theta, \sin \theta)$ and $Q(\cos(\alpha - \theta), \sin(\alpha - \theta))$
 lie on the circle $x^2 + y^2 = 1$
 Clearly $\angle XOP = \theta$

$$\angle XOQ = \alpha - \theta$$

If OR be the bisector of angle POQ then

$$\angle ROQ = \frac{\theta - (\alpha - \theta)}{2} = \theta - \frac{\alpha}{2}$$



$$\therefore \angle xOR = \left(\theta - \frac{\alpha}{2} \right) - (\theta - \alpha) = \frac{\alpha}{2}$$

$\Rightarrow$ Slope of line OR is $\tan \frac{\alpha}{2}$.

Thus Q is obtained from P by reflection in the line through origin with slope $\tan \frac{\alpha}{2}$.

10. Intersection point of $y = 0$ with first line is $B(-p, 0)$
 Intersection point of $y = 0$ with second line is $A(-q, 0)$
 Intersection point of the two lines is $C(pq, (p+1)(q+1))$
 Altitude from C to AB is $x = pq$

$$\text{Altitude from B to AC is } y = -\frac{q}{1+q}(x+p)$$

Solving these two we get $x = pq$ and $y = -pq$
 $\therefore$ locus of orthocentre is $x + y = 0$.

11. $\tan B + \tan C = \text{a constant}$
 $\Rightarrow 2R \cos B \cos C = \text{a constant}$
 $\Rightarrow$ Locus of orthocentre is straight line
 Since distance of orthocentre from BC is

$$2R \cos B \cos C$$

12. Let A be (x, y) and orthocentre be $(\alpha, 3 - \alpha)$

$$\therefore BO \perp BC \Rightarrow \frac{y-7}{x-4} \cdot \frac{\alpha}{1-\alpha} = -1 \dots \dots \dots (1)$$

$$\therefore AO \perp BC \Rightarrow \frac{y-3+\alpha}{x-\alpha} \times \frac{4}{3} = -1 \dots \dots \dots (2)$$

Eliminating α from (1) and (2) we get (3)

13. The given lines are inclined at 60° . The line $ax + by = 1$ is an acute angle bisector

STRAIGHT LINES

$$\cos \alpha + \sin(30^\circ - \alpha)$$

$$\frac{a}{-(\sin \alpha + \cos(30^\circ - \alpha))} = \frac{1}{2P} = \frac{\sqrt{a^2 + b^2}}{\sqrt{3}}$$

$$= a^2 + b^2 = \frac{3}{4P^2}$$

14. P lies on perpendicular bisector of BC and at

$$\text{a distance of } \frac{\sqrt{3}}{2} BC = \sqrt{15} \text{ units.}$$

from the midpoint of BC

$\therefore$ P can be $(\sqrt{3}, 2 - 2\sqrt{3})$ or $(-\sqrt{3}, 2 + 2\sqrt{3})$

But A and P should lie on the same side of BC

$\therefore$ P is $(-\sqrt{3}, 2 + 2\sqrt{3})$

15. Vertex A is mirror image of excentre about

$$x + y - 2 = 0, \therefore (\alpha, \beta) \text{ is } (6, 0)$$

16. Let the line be $\frac{x}{a} + \frac{y}{b} = 1 \Rightarrow b = 2a$

Let C be (α, α) , D be (h, k)

$$\therefore h + \alpha = a \quad k + \alpha = b$$

$$\therefore ABCD \text{ is a rhombus } \frac{\alpha - k}{\alpha - h} \times \frac{b}{-a} = -1$$

Locus of (h, k) is $x + y = 0$

17. The lines are $y = 2x$; $y = 3x$

Let $\angle OPB = \alpha = \angle OAB$ and $\angle AOB = \theta$

$$\tan \theta = \frac{1}{7} \text{ and } AB = OP \sin \theta$$

$$= 5 \times \frac{1}{5\sqrt{2}}$$

18. Equation of any line through $(2, 3)$ is $y = mx - 2m + 3$

$$\text{Area} = \frac{1}{2} \left(\frac{2m-3}{m} \right) (3-2m) = \pm 12$$

taking + sign we get $(2m+3)^2 = 0$

taking - sign we get

$$4m^2 - 36m + 9 = 0 \Rightarrow 3 \text{ St. lines are possible}$$

19. Let the coordinates of C be $(1, c)$, A = $(0, 0)$, B (x, y)

$$m_2 = \frac{c-y}{1-x}; m_2 = \frac{c-m_1x}{1-x}$$

JEE ADVANCED - VOL - IV

$$c = (m_1 - m_2)x + m_2 \dots\dots (1)$$

$$\text{now area of } \Delta ABC = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ x & m_1x & 1 \\ 1 & c & 1 \end{vmatrix}$$

$$= \frac{1}{2} (m_1 - m_2)(x - x^2) \quad [\because x > x^2 \text{ in } (0, 1)]$$

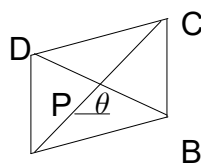
$$f(x)_{\max} = \frac{1}{8} \text{ when } x = \frac{1}{2}$$

MULTIANSWER QUESTIONS

$$20. \left(\frac{a+b}{2}, \frac{a+b}{2} \right) \text{ and } \left(\frac{a-b}{2}, \frac{b-a}{2} \right)$$

21. Possible coordinates are $(a, 0)$, $(0, -a)$

$$\left(\frac{a\sqrt{3}}{2}, -\frac{a}{2} \right), \left(-\frac{a\sqrt{3}}{2}, \frac{a}{2} \right)$$



22.

A

B = $(1, -3)$, D = $(0, 4)$, P = $(1/2, 1/2)$

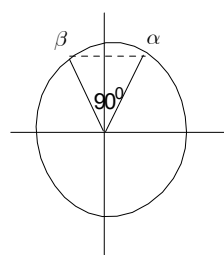
$$\tan \theta = \frac{1}{7}, \sin \theta = \frac{1}{5\sqrt{2}}, \cos \theta = \frac{7}{5\sqrt{2}}$$

$$AP = \frac{5}{2} \sqrt{2} \cot 30^\circ = \frac{5}{2} \sqrt{2} \sqrt{3}$$

$$A = \left(\frac{1}{2} \pm AP \cos \theta, \frac{1}{2} \pm AP \sin \theta \right)$$

$$23. (2a + 3b - c)(3a + 4b - 2c) = 0$$

24.



$$\alpha - \beta = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}$$

25. The equation of side AB is

$$L(x, y) = \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix}$$

$$\therefore L(0, 0) = x_1 y_2 - x_2 y_1$$

$$\text{and } L(x_3, y_3) = \begin{vmatrix} x_2 & y_3 & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} lx_1 + mx_2 & ly_1 + my_2 & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & 1-1-m \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} (R_1 - lR_2 - mR_3)$$

$$= (1-1-m)(x_1 y_2 - x_2 y_1)$$

$$\therefore L(0, 0) \cdot L(x_3, y_3)$$

$$= (1-1-m)(x_1 y_2 - x_2 y_1)^2 > 0$$

$\therefore$ Origin and C lies on same side of AB

Clearly, Origin lies inside $\triangle ABC$

26. use parametric form

COMPREHENSION QUESTIONS

PASSAGE : 1 (27 TO 29)

$$A = (1, 3), C = \left(-\frac{2}{5}, -\frac{2}{5}\right) \text{ Let } B = (\alpha, 2 - \alpha)$$

Lies on $x + y = 2$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\left| \frac{2 + \frac{2}{5} - \alpha}{\alpha + \frac{2}{5}} + 1 \right| = \left| \frac{1 + \alpha}{1 - \alpha} + 1 \right| \Rightarrow \alpha = -\frac{5}{2}$$

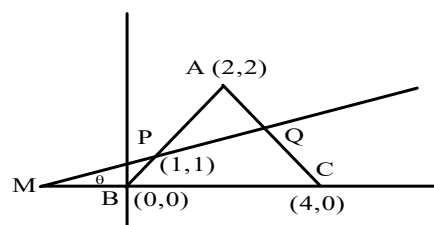
$$\therefore B = \left(-\frac{5}{2}, \frac{9}{2}\right)$$

Equation of BC is $7x + 3y + 4 = 0$

Equation of AB is $3x + 7y - 24 = 0$

PASSAGE : 2 (30 TO 32)

Equation of line PM



$$y - 1 = \tan \theta (x - 1)$$

Intersection point Q of AC and MP

$$4 - x - 1 = \tan \theta (x - 1)$$

$$\Rightarrow Q \equiv \left(\frac{3 + \tan \theta}{1 + \tan \theta}, \frac{1 + 3 \tan \theta}{1 + \tan \theta} \right)$$

Area of $\triangle APQ$ = modulus of

$$\frac{1}{2} \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ \frac{3 + \tan \theta}{1 + \tan \theta} & \frac{1 + 3 \tan \theta}{1 + \tan \theta} & 1 \end{vmatrix} = \frac{1 - \tan \theta}{1 + \tan \theta}$$

30. Area of quadrilateral BPQC

$$A = \frac{1}{2} \times 4 \times 2 - \left(\frac{1 - \tan \theta}{1 + \tan \theta} \right)$$

$$= \frac{3 + 5 \tan \theta}{1 + \tan \theta}$$

$$31. A = 5 - \frac{2}{1 + \tan \theta}$$

$$\theta = \left(0, \frac{\pi}{4} \right) \Rightarrow 1 + \tan \theta \in (1, 2)$$

$$\Rightarrow A \in (3, 4)$$

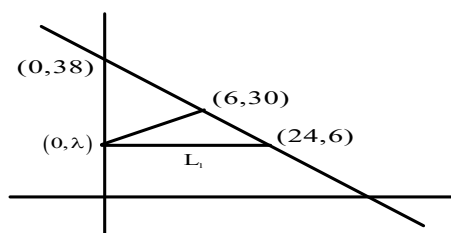
$$32. (PQ)^2 = \left(\frac{3 + \tan \theta}{1 + \tan \theta} - 1 \right)^2 + \left(\frac{1 + 3 \tan \theta}{1 + \tan \theta} - 1 \right)^2$$

$$= \frac{4(1 + \tan^2 \theta)}{1 + \tan^2 \theta + 2 \tan \theta} = \frac{4}{1 + \sin 2\theta}$$

$$\Rightarrow (PQ)^2 \in (2, 4) \Rightarrow PQ \in (\sqrt{2}, 2)$$

PASSAGE : III (33 TO 35)

STRAIGHT LINES



$$M_{L_1} = \frac{\lambda - 6}{-24}, M_{L_2} = \frac{\lambda - 30}{-6}$$

$$\tan \theta = \frac{M_{L_2} - M_{L_1}}{1 + M_{L_2} M_{L_1}} = \frac{\frac{30 - \lambda}{6} - \frac{6 - \lambda}{24}}{1 + \left(\frac{30 - \lambda}{6}\right)\left(\frac{6 - \lambda}{24}\right)}$$

$$= \frac{(38 - \lambda)18}{(\lambda - 18)^2}$$

$$\frac{d}{d\lambda}(\tan \theta) = \frac{18(\lambda - 18)(\lambda - 58)}{(\lambda - 18)^4}$$

33. $\tan \theta$ is maximum net $\lambda = 18$

34. $(\tan \theta)_{\max} \rightarrow \infty \Rightarrow \theta = \frac{\pi}{2}$

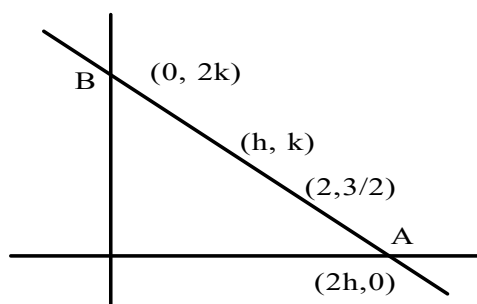
35. $\tan \theta = \frac{m_{L_1} - m_{L_2}}{1 + m_{L_1} m_{L_2}} = \frac{(\lambda - 38)18}{(\lambda - 18)^2}$

$$\frac{d}{d\lambda}(\tan \theta) = \frac{18(18 - \lambda)(\lambda - 58)}{(\lambda - 18)^4}$$

$\tan \theta$ is minimum at $x = 58$

PASSAGE : IV (36 TO 38)

36. Intersection point of line is



$$\left(2, \frac{3}{2}\right)$$

Equation of AB is

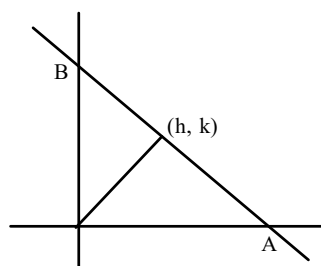
$$\frac{x}{2k} + \frac{y}{2k} = 1$$

$$\Rightarrow \frac{1}{h} + \frac{3}{4k} = 1$$

JEE ADVANCED - VOL - IV

$\Rightarrow 4k + 3h = 4hk \Rightarrow 3x + 4y - 4xy = 0$

37. Equation of AB is



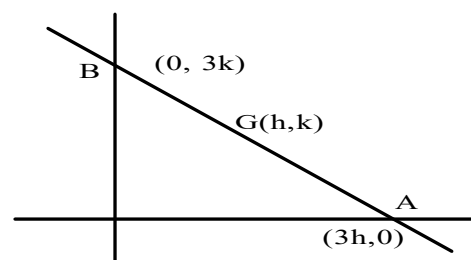
$$hx + ky = h^2 + k^2$$

Put, $\left(2, \frac{3}{2}\right)$

$$2h + \frac{3k}{2} = h^2 + k^2$$

$$\Rightarrow 2(x^2 + y^2) - 4x - 3y = 0$$

38. Equation of AB is $\frac{x}{3h} + \frac{y}{3k} = 1$



Put $\left(2, \frac{3}{2}\right)$

$$\frac{2}{3h} + \frac{1}{2k} = 1 \Rightarrow 4k + 3h = 6hk$$

PASSAGE : V (39 TO 41)

$$a + b + \sqrt{a^2 + b^2} = \frac{1}{2}ab$$

$$(2a + 2b - ab)^2 = 4(a^2 + b^2)$$

$$8ab + a^2 + b^2 - 4ab|a + b| = 0$$

$$(a - 4)(b - 4) = 8 = 2 \times 4, 1 \times 8$$

$$(a, b) = (6, 8), (5, 12)$$

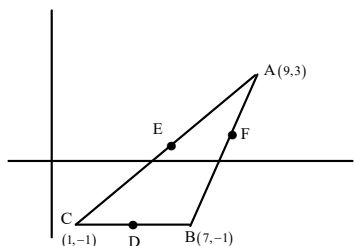
39. No of ordered pair $(a, b) = 2$

40. For maximum perimeter
 $a = b, b = 12 \Rightarrow$ Maximum perimeter
 $= 5 + 12 + 13 = 30$

41. For minimum area, $a = 6, b = 8$

$$\text{Minimum area} = \frac{1}{2} \times 6 \times 8 = 24$$

PASSAGE : VI (42 TO 44)



42. Area of triangle DEF = $\frac{1}{4}$ area of ABC
43. Equation of perpendicular bisector of AC is
 $y - 1 = (-2)(x - 5)$
 $y + 2x = 11$
 Perpendicular bisector of BC is $x = 4$
 $\therefore$ circumcentre $\equiv (4, 3)$
 $\Rightarrow R = \sqrt{(4-1)^2 + (3+1)^2} = 5$
 $a + b + R = 4 + 3 + 5 = 12$
44. O $\equiv$ Circumcentre, G $\equiv$ Centroid
 $H \equiv \left(3 \cdot \frac{17}{3} - 2 \cdot 4, 3 \left(\frac{1}{3} \right) - 2(3) \right) \equiv (9, -5)$

PASSAGE : VII

45. $\perp$ distance from C_1 to AC = $\frac{|0+0-1|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$
 D is the foot of the perpendicular of C_1 onto AC
 D is also the midpoint of AC
 Coordinates of D is $(1/2, 1/2)$
 Coordinates of B is $(h, 1-h)$
 $BD = 3C_1D = 3/\sqrt{2}$
 In an equilateral triangle height = $BD = \frac{\sqrt{3}}{2}a$ (a is the side of the triangle)
 $\Rightarrow a = \sqrt{6}$
 $BA = \sqrt{(h+1)^2 + (1-h+1)^2} = \sqrt{6}$
 solving we have $h = \frac{1-\sqrt{3}}{2}$ or $\frac{1+\sqrt{3}}{2}$

$$\frac{x_1x_2 + x_2x_3 + x_3x_1}{\left(\frac{1-\sqrt{3}}{2}\right)\left(\frac{1+\sqrt{3}}{2}\right) + -1\left(\frac{1-\sqrt{3}}{2} + \frac{1+\sqrt{3}}{2}\right)} =$$

$$= \frac{1-3}{4} - 1 = -\frac{3}{2}$$
46. $\angle BCA = \angle BPA = 60^\circ$ (angles in the same segment)

STRAIGHT LINES

Let $\angle CAP = \theta$ and circumradius of $\triangle ABC = R$

$$PA = 2R \sin(60 - \theta) \text{ (sine rule in } \triangle APB)$$

$$PC = 2R \sin \theta \text{ (sine rule in } \triangle BPC)$$

$$PB = 2R \sin(60 + \theta) \text{ (sine rule in } \triangle APB)$$

we have $PB = PA + PC$

$$\Rightarrow (PA + PB + PC)_{\max} = (2PB)_{\max}$$

$$= 2 \times 2R = 4R = 4R = \frac{\sqrt{6}}{2 \sin 60}$$

PB is maximum when it is the diameter hence P is $(1, 1)$

47. Observe that $P^1A + P^1C \geq AC$ (Triangle inequality)

$$P^1B + P^1P \geq BP$$

$$\text{So } P^1A + P^1B + P^1C + P^1P \geq AC + BP$$

$$\geq \sqrt{6} + 2\sqrt{2}$$

$$\sqrt{6} + \sqrt{8}$$

$$\text{Hence If } P^1A + P^1B + P^1C + P^1P \geq \sqrt{6} + \sqrt{8}$$

P^1 can only be $(1/2, 1/2)$

PASSAGE-VIII

48. Clearly the possible point will be the intersection of the circle with centre vertex and radius as side of the square and the line which is perpendicular bisector of the sides. From this 8 different position of P can be find and one point will be the centre of the square.
49. Clearly for all the different position of 'P' except origin there will be one equilateral triangle.
50. Different position of P except origin will be $(0, -\sqrt{3}-1), (\sqrt{3}+1, 0), (\sqrt{3}-1, 0), (-\sqrt{3}-1, 0)$
 $(0, -\sqrt{3}+1), (0, -\sqrt{3}+1), (\sqrt{3}-1, 0)$ and $(-\sqrt{3}-1, 0)$

PASSAGE - IX

51.52,53) Let $A = (1, \lambda)$, Equation of AB, AC are

$$y - \lambda = \pm 2(x - 1)$$

$$B = (1 - \lambda, -\lambda), C = \left(\frac{5+3\lambda}{5}, -\frac{\lambda}{5} \right)$$

slope of $BC = \frac{1}{2}$ consequently $BC \perp AC$

MATRIX MATCH QUESTIONS

STRAIGHT LINES

JEE ADVANCED - VOL - IV

$$54. \quad A) \begin{vmatrix} 1 & -2 & -6 \\ 3 & 1 & -4 \\ \lambda & 4 & \lambda^2 \end{vmatrix} = 0$$

$$\Rightarrow 1(\lambda^2 + 16) + 2(3\lambda^2 + 4\lambda) - 6(12 - \lambda) = 0$$

$$\Rightarrow 7\lambda^2 + 14\lambda - 56 = 0 \Rightarrow \lambda^2 + 2\lambda - 8 = 0$$

$$\Rightarrow \lambda = 2, -4$$

$$B) \text{ Collinear } \Rightarrow \frac{2}{\lambda} = 2\lambda - 3 \Rightarrow \lambda = -\frac{1}{2}, 2$$

$$C) \text{ Point of intersection of } x - y + 1 = 0 \text{ and } 3x + y - 5 = 0 \text{ is } (1, 2). (1, 2) \text{ lies on } x + y - 1 - \lambda = 0 \Rightarrow \lambda = 2$$

$$D) \text{ Perpendicular distance from } (1, -2) \text{ and } (3, 4) \text{ to } y - x - 1 + \lambda = 0 \text{ are equal}$$

$$\Rightarrow \frac{|\lambda - 4|}{\sqrt{2}} = \frac{|\lambda|}{\sqrt{2}} \Rightarrow \lambda - 4 = -\lambda$$

$$\Rightarrow \lambda = 2$$

$$55. \quad x + y + 1 = 0, \text{ Arbitrary point on the above line as } (\alpha, -\alpha - 1)$$

$$\text{Image of } (\alpha, -\alpha - 1) \text{ on } 2x + y + 1 = 0 \text{ be } (h, k)$$

$$\Rightarrow \frac{h - \alpha}{2} = \frac{k - (-\alpha - 1)}{1} = \frac{-2(2\alpha + 1 - \alpha - 1)}{2^2 + 1}$$

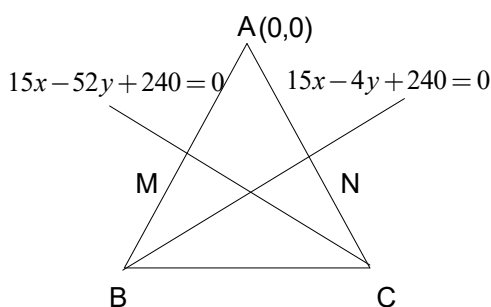
$$= \frac{-2\alpha}{5}$$

$$\Rightarrow (h, k) = \left(\frac{\alpha}{5}, \frac{-7\alpha}{5} - 1 \right)$$

$$\text{Locus of } (h, k) = 7x + y + 1 = 0$$

$$B, C, D \text{ similarly}$$

56.



$$B = \left(x_1, \frac{15x_1 - 240}{4} \right)$$

$$\text{and } C = \left(x_2, \frac{15x_2 + 240}{52} \right)$$

$$M = \left(\frac{x_1}{2}, \frac{15x_2 - 240}{8} \right)$$

$$N = \left(\frac{x_2}{2}, \frac{15x_2 + 240}{104} \right)$$

57. Conceptual.

58. The line joining the orthocentre and circumcentre is parallel to the side BC. Distance from orthocentre to the vertex A is $2R \cos A$ and the distance from orthocentre to side BC is $2R \cos B \cos C$.

INTEGER QUESTIONS

59. Let the lines represented by the given equation be $y = mx$ and $y = 4mx$, then

$$5m = \frac{-2h}{72}, 4m^2 = \frac{36}{77}$$

$$\Rightarrow h^2 = 4050$$

60. Any line through A(2, 5) is $\frac{x-2}{\cos \theta} = \frac{y-5}{\sin \theta} = r$

Any point on the line is

$$R = (2 + r \cos \theta, 5 + r \sin \theta)$$

Two lines of $2x^2 - 5xy + 2y^2 = 0$ are

$$x = 2y \text{ and } y = 2x$$

Put R in $x = 2y$ then $r = AP$, we get

$$2 + r \cos \theta = 2(5 + r \sin \theta)$$

$$\Rightarrow \frac{1}{r} = \frac{\cos \theta - 2 \sin \theta}{8} = \frac{1}{AP}$$

$$\text{Similarly, } \frac{1}{AQ} = \sin \theta - 2 \cos \theta$$

Since AP, AR, AQ are in H.P.

$$\Rightarrow \frac{2}{r} = \frac{1}{AP} + \frac{1}{AQ}$$

$$= \frac{17 \cos \theta - 10 \sin \theta}{8} \quad (1)$$

$$\text{Let } R = (h, k) \Rightarrow \cos \theta = \frac{h-2}{r}, \sin \theta = \frac{k-5}{r}$$

$$\text{From (1), } \frac{2}{r} = \frac{17h - 34 - 10k + 50}{8r}$$

$$\Rightarrow 17h - 10k = 0$$

$$\text{Locus of R is } 17x - 10y = 0, \quad \therefore \lambda = 17$$

Now, $\frac{\lambda+1}{6} = \frac{18}{6} = 3$

61. Given, $\frac{a\left(\sum_{n=1}^{2013} n\right) + b\left(\sum_{n=1}^{2013} n^2\right) + c\sum_{n=1}^{2013} (1)}{\sqrt{a^2 + b^2}} = 0$

$$a\left(\frac{2013 \times 2014}{2}\right) + b\left(\frac{2013 \times 2014 \times 4027}{6}\right) + 2013c = 0$$

$$\Rightarrow a(1007) + b\left(\frac{1007 \times 4027}{3}\right) + c = 0$$

$$\alpha = 1007, \beta = \frac{1007 \times 4027}{3}$$

$$\begin{aligned} \text{Now, } \frac{1}{\alpha}\left(\beta - \frac{1007}{3}\right) &= \frac{1}{1007}\left(\frac{1007}{3}(4026)\right) \\ &= 1342 \\ &= abcd \\ b &= 3, d = 2 \\ b + d &= 3 + 2 = 5 \end{aligned}$$

62. Let the equation of L_2 be $L_1 + \lambda L = 0$

$$\Rightarrow (1+\lambda)x + (2+\lambda)y + (3+\lambda) = 0$$

Slope of L_2 , L and L_1 are

$$-\frac{1+\lambda}{2+\lambda}, -1, -\frac{1}{2}$$

Since L is the bisector of the angle between L_1 and L_2

$$\therefore \frac{-\frac{1+\lambda}{2+\lambda} + 1}{1 + \frac{1+\lambda}{2+\lambda}} = \frac{-1 + \frac{1}{2}}{1 + \frac{1}{2}} \Rightarrow \lambda = -3$$

So the equation of L_2 is $y + 2x = 0 \Rightarrow m = -2$

63. Making the equation of the circle homogenous with the help of the equation of line, we get

$$\text{Coefficient of } x^2 + \text{coefficient of } y^2 = 0$$

$$\Rightarrow c^2 = 5386$$

64. Let $y = mx$ be one of the lines through the origin. Since the triangle is isosceles right angled, this

line makes an angle of $\pm \frac{\pi}{4}$ with $2x + 3y = 6$

whose slope is $-\frac{2}{3}$

$$\text{So, } \tan\left(\pm \frac{\pi}{4}\right) = \frac{m + \frac{2}{3}}{1 + m\left(-\frac{2}{3}\right)} = \frac{3m+2}{3-2m}$$

$$\therefore m = y/m$$

$$\therefore 5x^2 - 24xy - 5y^2 = 0$$

$$\text{Comparing with } ax^2 - 2hxy - ay^2 = 0$$

$$a = 5 \text{ and } h = 12$$

65. Equation of OA is $y = \frac{3}{4}x$

$$\text{Equation of BC is } y = \frac{3}{4}x + l$$

$$\text{Equation of OC is } y = \frac{12}{5}x$$

$$\text{Equation of AB is } y = \frac{12}{5}x + m$$

We must have, $m < 0, l > 0$

($\because$ Y intercept of AB is negative and that of BC > 0)

Now OABC is a rhombus,

Distance between OA and BC

= Distance between OC and AB

$$\Rightarrow \left|\frac{4l}{5}\right| = \left|\frac{5m}{13}\right| \Rightarrow m = \frac{-52}{25}l$$

$$\therefore B = \left(\frac{20(l-m)}{33}, \frac{16l-5m}{11}\right)$$

$$\text{Given } OB^2 = 144$$

$$\Rightarrow \frac{400(1-m)^2}{33 \times 33} + \frac{(16l-5m)^2}{121} = 144$$

$$\text{On putting } m = \frac{-52}{25}l \text{ and solving for } l, \text{ we get}$$

$$l = \frac{45}{\sqrt{130}} \Rightarrow m = -\frac{468}{5\sqrt{130}}$$

$$\therefore a = 130$$

STRAIGHT LINES

$$\therefore \frac{a-40}{10} = \frac{90}{10} = 9$$

66. **Slope of AB x slope of AC = -1** $\therefore \angle A = 90^\circ$

Let AC = P; BC = P+50 and AB = a

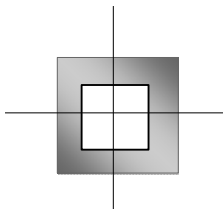
$$\therefore a^2 = (P+50)^2 - P^2$$

$$a = 10\sqrt{P+25}$$

$$\therefore S = 60 + 11 + 61 = 132$$

$$as = 7920 = 1930 \times 4$$

67. $\frac{|x+y|}{\sqrt{2}} + \frac{|y-x|}{\sqrt{2}} \in [2, 4]$



if $y \geq x$ and $y+x \geq 0$

$$\Rightarrow y \in [\sqrt{2}, 2\sqrt{2}]$$

if $y \leq x$ and $y+x \leq 0$

$$\Rightarrow y \in [-2\sqrt{2}, -\sqrt{2}]$$

if $y \geq x$ and $y+x \leq 0$

$$\Rightarrow x \in [-2\sqrt{2}, -\sqrt{2}]$$

if $y \leq x$ and $y+x \geq 0$

$$\Rightarrow x \in [\sqrt{2}, 2\sqrt{2}]$$

area of region

$$= (4\sqrt{2})^2 - (2\sqrt{2})^2 = 24$$

68. Ortho centre of $\triangle ABC$ is in centre of

$\triangle DEF$

$\therefore$ In centre of $\triangle DEF$ is (10, 5) and A,B,C are (5, 10), (50, -5), (15, 30) taking any three of them we get a Triangle satisfying the conditions. Hence 4 such Triangles are possible.

69. The curve represents a point $\left(-\frac{1}{2}, -1\right)$ and the line $2x + y - 1 = 0$.